MEDICOOL/HX 10.0 P7

10 Kilo-Watt Gradient Water Heat Exchanger

User Manual

AIRSYS GROUP

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Symbols Used on the MEDICOOL/HX 10.0 P7

This symbol marks chapters and sections of this instruction manual which are particularly relevant to safety.

When attached to the unit, this symbol draws attention to the relevant section of the instruction manual.

This symbol indicates hazardous voltages may be present.

A NOTE is used to call particular attention to a step of a procedure, which, if not strictly followed, could result in damage to, or destruction of equipment.

REVISION HISTORY

Revision 0 07/31/03		
Name	Section No.	Action
Zhang-ping	2	Illustration 4—Add minimum clearance for unit
	3.3.1	Add power requirement
	3.4.1	Add the facility water requirement
Revision 1 10/15/03		
Name	Section No.	Action
Zhang-ping	2	Illustration 4—Add a note that gives the weight of the heat exchanger when full
	2	Illustration 4—Add location of power inlet port, show dimensions
	1.4	Table 4—show the net weight for empty and full conditions
	1.1	step 4--add a ramp built into the top of the crate that can be removed and placed under the casters so that the unit can be removed from the skid.
	1.2	table 1—remove 1m section of ½" ID rubber hose
		Remove any mention of coldhead compressor in this manual
	1.2	table 1—Divide the table based on the contents of the two shipping containers. Have the heat exchanger, control box key, user manual and warranty card in the first box and all the other items in the second box
	1.2	table 1—Add 9.1 meter power cable to list
	1.3	Operating conditions—Remove high altitude note
	3.2	Remove the the paragraph starting with "In the lower..." as well as Illustration 5 and 6. We will provide separate seismic mounting instructions with our special mounting kit
	3.5	Add a note that says "Important! Use only the coolant provided or GE part number 2297672
	4	step 3—Reword this to say "Connect power cord to appropriate terminal block in the GE system Power Distribution Unit"
revision 2 11/07/03		

Name	Section No.	Action
Zhang-ping	1.1	Add weight of each unit shipping box
	3.4.1	Add facility water cooling specifications
	3.4.2	change Pipe connection to hose connection
	5.1	remove the Siemens name from the display
	Appendix	Add piping diagram
	3.4.1	Increase height of shaded area to 40 l/min. Label shaded area as "admissible range"
	4	Before the step in which the pump is started, add a step to remove the right side cover, shut the pressure control valve (refer to Illustration 12) then open it two turns
	4	After the step in which the pump is started, add a step to open the pressure control valve until the outlet pressure gauge reads 0.45 Mpa then replace the right side cover
	5.3	Add a callout for the pressure control valve and the outlet pressure gauge
revision 3 02/04/04		
Name	Section No.	Action
yangzhen	1.2 table3	Add the Teflon tape 20mX25mmX0.1mm
	2	Modify the lable site of "facility water return"、"facility water supply"、"from gradient coil"、"to gradient coil"
revision 4 03/03/04		
Name	Section No.	Action
yangzhen	1.1	Add the shipping containers weight is 150kg(333lb)
	5.3	Add "Follow all applicable G.E.Lock-out,tag-out procedures."
	9	Add the table of the spare part
revision 5 03/09/04		
Name	Section No.	Action
yangzhen	Section4 7)	Add the "see illustion 12"
Revision 6 03/19/04		
Name	Section No.	Action
yangzhen	3.4.1	Add the value "180" of "maximum heat output to water"

Revision 7 06/01/04		
Name	Section No.	Action
yangzhen	Appendix B:pipediagram	Add a restrict valve and bapass valve at the outlet of the pump
Revision 8 07/21/04		
Name	Section No.	Action
yangzhen	after3.2.2	Add section 3.2.3
Revision 9 02/24/09		
Name	Section No.	Action
Jia Runyu	Section 4	Add Pump air bleed

Table 1 revision history

1. Section 1 – General Information

1.1. Unpacking

Your unit is transported in a special carton. the dimensions of the shipping containers(mm X mm X mm) :990X770X1300 and its weight is 150KG(3331b);the dimensions of the box for loose parts(mm X mm X mm):800X700X870 and its weight is 178.5kg(394 lb).

1. Insert a pry bar into the gap between top of crate with crate, Remove top of the crate and place in an area clear of the uncrating area. (See Illustration 1).

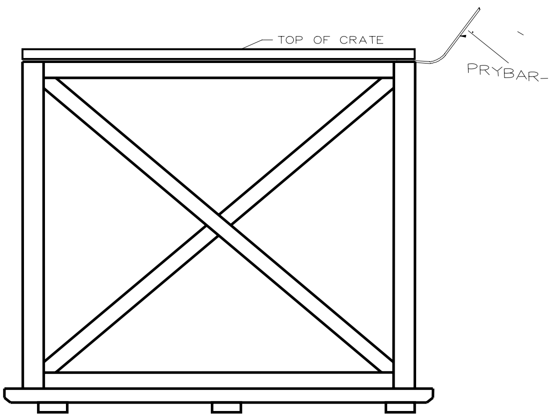


Illustration 1 remove the top of crate

Using a pry bar carefully remove the front, sides, and back of the crate and place in an area clear of the uncrating area.(see Illustration 2)

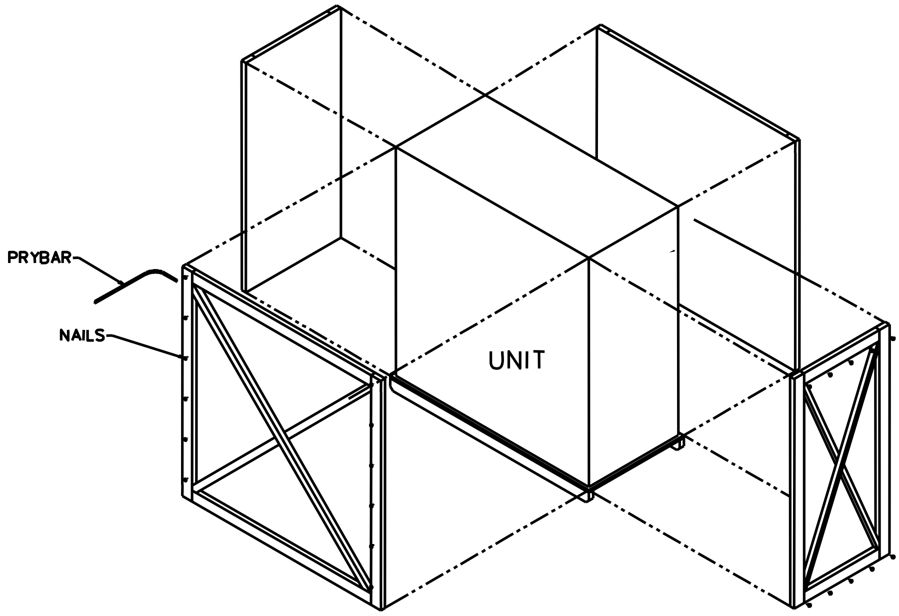


Illustration 2 remove other parts of crate

2. CAREFULLY remove plastic.
3. Inspect equipment for any damage. Contact Air-Sys Group(Air Sys can be contacted at the phone number at the bottom of the page) and the transport company as soon as possible and report any discovered damage.
4. push the unit onto a ramp and roll it down the ramp which is built into the lid of the shipping container to where you want to place it. (see Illustration 3).

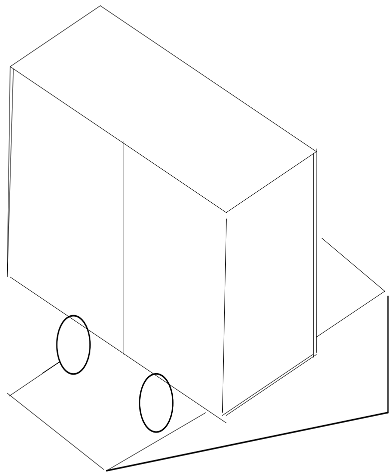


Illustration 3 lift the unit

Retain the carton and all packing materials until the unit is completely assembled and working properly. Set up and run the unit immediately to confirm proper operation. If the unit is damaged or does not operate properly, contact the transportation company, file a damage claim.

1.2. Package Contents (MEDICOOL/HX 10.0 P7)

Item	Quantity
Gradient Water Heat Exchanger	1
Key to control section	4
Fuse(0.5A)	1
User Manual	1
Warranty Card	1

Table 2 CONTENTS OF SHIPPING BOX

Item	Quantity
19.1mm (3/4”) hose barbs	4
19.1mm (3/4”) hose clamps	8
19.1mm(3/4”) male quick disconnect fittings	2
19.1mm(3/4”) female quick disconnect fittings	2
30.5m (100ft) lengths of 19.1mm (3/4”) flexible rubber hose	2
Terminals with size of 1.0mm^2(1.55×10^{-3}in^2), 1.5mm^2(2.33×10^{-3}in^2), 6mm^2(9.3×10^{-3}in^2)	6 for each
Mixture of propylene glycol and purified water (50/50 mixture)	0.81m^3(21.5 gal)
3 conductor,12 AWG power cable	9.1m(30ft)
Funnel	1
Teflon tape 20mX25mmX0.1mm	1 roll
Wrench	1

Table 3 CONTENTS OF LOOSE PART BOX

1.3. Description

The common heat exchanger is a single-loop device used to circulate water through the gradient coil with the purpose of removing heat generated during operation and transferring this heat to the site chilled water system. It consists of a heat exchanger unit, coolant reservoir and pump contained within an enclosure. The heat exchanger will maintain the temperature of a propylene glycol/water (PGW) mixture at an adjustable setpoint which will be preset to 20°C±1°C(68°F±33.8°F) with an adjustable range of

15°C(59°F) to 25°C(77°F).

All models have a microprocessor controller, by digital set/read and readout in °C. From conception, this unit has been designed for a long life and an ease of use. The heat exchanger has a MTBF (Mean Time Between Failures) of 20,000 hours in the operating environment.

- ## Environmental conditions:

- **Non-operating**

Ambient Temperature -34°C (-29.2°F) to +55°C (131°F)

Altitude 120m (400 ft) below to 3352m (11,000 ft) above sea level

Humidity Up to 100% RH

Magnetic Field 30 Gauss

- **Operating**

Ambient Temperature 5°C (41°F) to +40°C (104°F)

Altitude 30m (100 ft) below to 1524m (5,000 ft) above sea level

Humidity 10% to 90% RH with minimum dew point at 20°C(68°F) an maximum wet bulb temperature at 30°C(86°F)

Magnetic Field 30 Gauss

1.4. Specifications

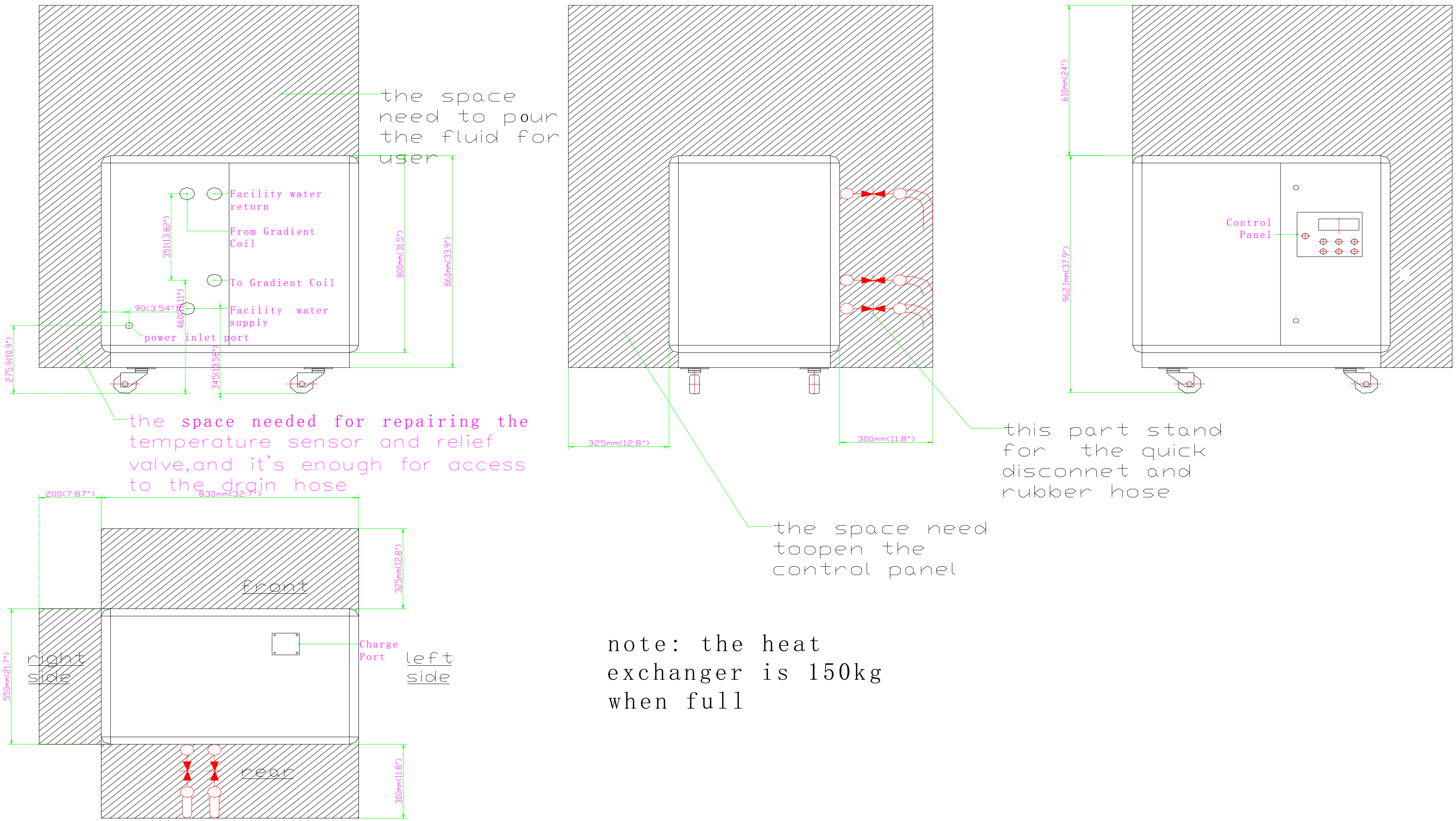
		Hot Side	Cold Side
Cooling Capacity (1)	KW	10.0	10.0
Plate heat exchanger			
Water Flow (1)	l/min(gal/min)	23.1(6.1)	Min=19(5.0);max=40(10.6)
Water Resistance (1)	KPa(lbf/in^2)	112(16.2)	Min=80(11.6);max=120(17.4)
Water Content	l(gal)	2.1(0.6)	2.1(0.6)
Temperature Control Methods		P.I.D	
Water Pump	n.	1	
Head	m	50/72(50Hz/60Hz)	
Power	KW(Hp)	0.8/1.1(1.1/1.5)	
Current	A	9@50Hz/13@60Hz	
Coolant		mixture of propylene glycol and purified water (50/50 mixture)	

Noise		dB(A)	69.0
Dimensions			
Length		mm(in)	830(32.7)
Width		mm(in)	550(21.7)
Height		mm(in)	980(38.6)
Net Weight	empty	Kg(lb)	120(265)
	full	Kg(lb)	150(331)
1-rated conditions: ambient temperature:21℃(69.8°F);supply water temperature:15℃(59°F);return water temperature:20℃(68°F)			

Table 4 SPECIFICATIONS

2. Section 2 –Unit Overall Dimensions & Minimum Service Clearances

Illustration 4 unit overall dimension



3. Section 3 – Hook up

Before proceeding, be sure all power is off

3.1. Installation Tools

Tool	Spec	Quantity	Usage
Line Type Screwdriver	50X2.5	1	Wiring
Line Type Screwdriver	65X5	1	Piping, Wiring
Spanner	300#	2	Piping

Table 5 INSTALLATION TOOLS

3.2. Location

Locate the heat exchanger on a strong, level surface or slightly inclined surface(not exceed 5° incline). Ensure easy access to the top cover. The wheels can be locked to keep the chiller in place while in use.

Note: The MEDICOOL/HX should be located above/below gradient coil within $\pm$5m(16ft).

If possible, be close to the power distribution panel.

3.3. Electrical Connections

3.3.1. Power requirements

ITEM	REQUIREMENT
Wire(customer supplied)	12 AWG(3.3mm^2)
Power Consumption	0.75Kw(maximum)
Voltage Requirements	120 VAC@ 50 Hz 120 VAC@ 60 Hz
Circuit Size	12AMP(maximum)

Table 6 Power Requirements

3.3.2. Electrical connections

Open the door switch on the control panel and remove screws from right cover. There is a connector with 3 couples of binding post. Put the power (120 VAC 50/60Hz) line through the power inlet port on the rear cover and electric box. Connect the “Live” to the post with marked wire of “100” (the 2nd from the left), and the “Neutral” to “103” (the 3rd from the left). Connect the “Ground” to the “PE”.

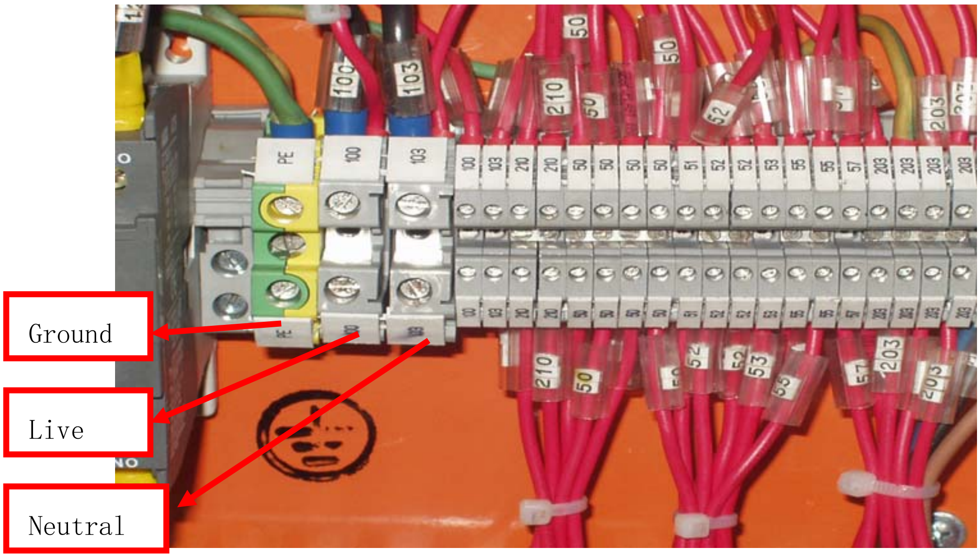


Illustration 5 Electrical connection

3.3.3. Power Connection to PDU from Heat Exchange

Remove AC panel from Gradient/PDU cabinet. See Illustration 6.

Illustration 6

Install power cable through AC panel. Make following connections:

a. Blue wire to location 21

b. White wire to Neutral

c. Green/yellow wire to Ground. See Illustration 7.

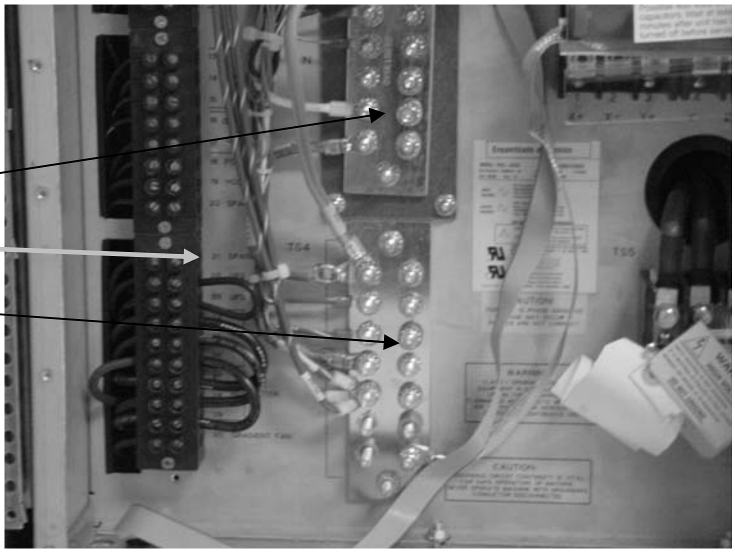


Illustration 7

Label bottom most breaker in the front of the PDU Heat Exchanger.

Re-connect AC panel to Gradient/PDU cabinet.

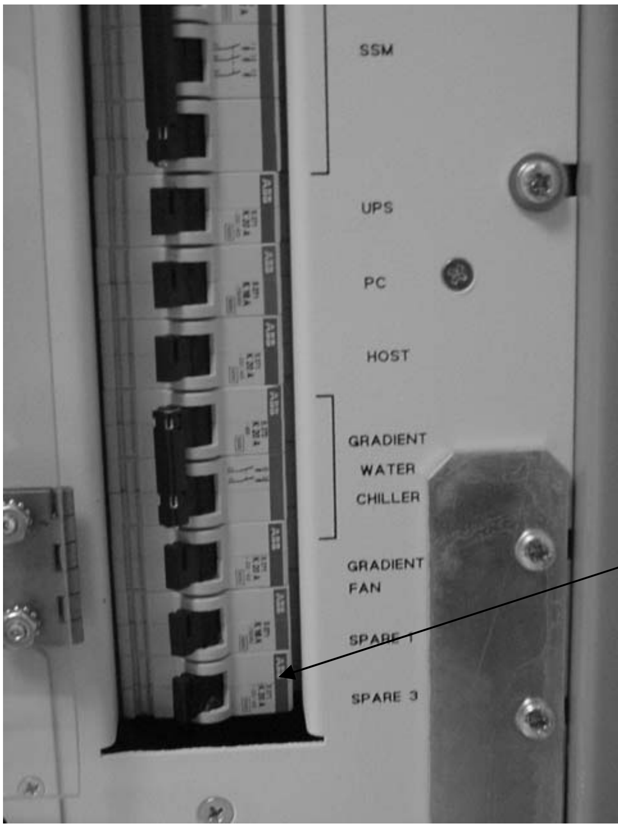


Illustration 8

3.4. Hose Connections

3.4.1. Facility input water requirements

Facility input water must meet all requirement with the following constraints:

Composition (water or water/glycol mix %)	50/50
Maximum heat output to water	180 Watt
Maximum heat output to air (if any)	0
Pressure Drop at minimum flowrate	120 KPa (17.4 lbf/in^2)
Pressure Drop at maximum flowrate	80 KPa (11.6 lbf/in^2)
Inlet pressure	112 KPa (16.21bf/in^2)
pH	6.5-8.2
Hardness	Less than 200 ppm of calcium carbonate
Suspended matter	Less than 10 mg per liter, less than 150 micron particle size
Facility water supply flow/temperature	Can be anywhere with shaded area of illustration 6

Table 7 facility water requirement

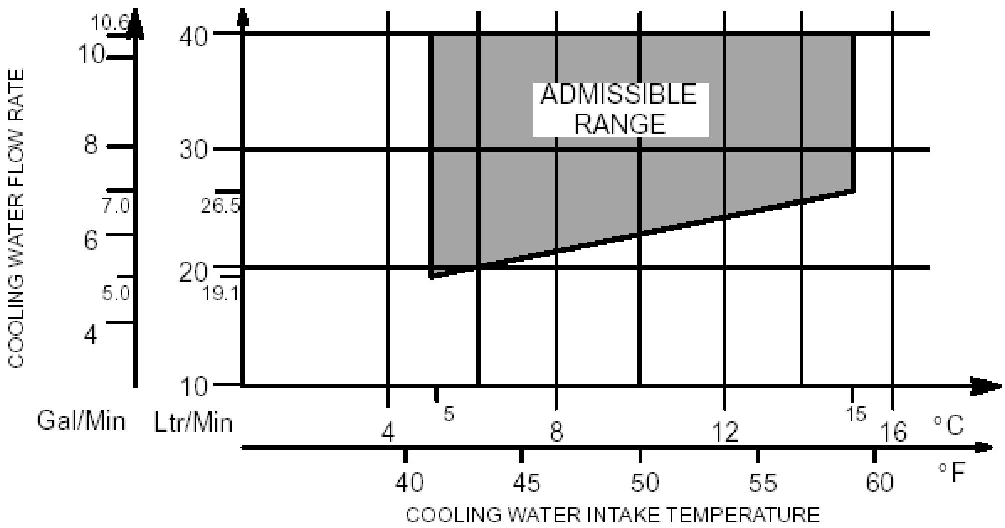
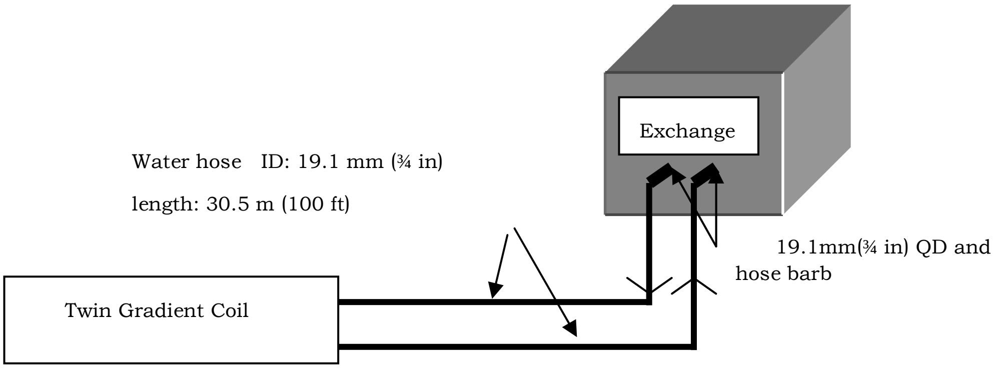


Illustration 9 acceptable range for facility water

3.4.2. Hose connection

Before connecting the water hose with the heat exchanger, check the water hose thoroughly to make sure there is no machining residuals or dirt inside the pipe. Check the hydraulic connections



to locate any water leaks.

Illustration 10 connect with Twin Grandient Coil

The process fluid inlet and outlet connections, located on the rear side, are 3/4" corrosion-resistant, metallic, double-shutoff type quick disconnect with 3/4" hose barb and hose clamp(see illustration 8).

Illustration 11 quick disconnection

3.5. Coolant Fill

Note: Important! Use only the coolant provided or GE part number 2297672

Remove Screws from coolant charge port cover on the top of unit, put funnel in the coolant charge port. Fill the reservoir until the liquid level reaches "coolant charge level" line . Close the top cover before operating.

Illustration 12 Coolant fill

4. Section 4 –Start-up

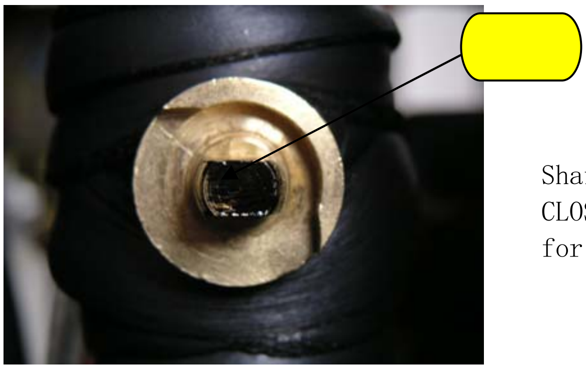
Start-up the MEDlCOOL/HX by the following steps:

Note: Before first starting the chiller, please do check electrical control system include control panel, all junction box inside or outside chiller for burned or frayed wiring and loose connections and do the insulation test of the field installing power input line used the megohm meter.

50Hz/60Hz Configuration:

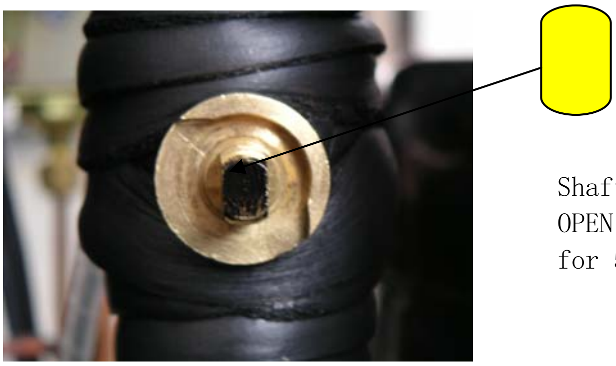
remove the right side cover and open the pressure control valve, a valve without bonnet can be seen in the outlet pipe of pump, which has two positions (open/close) and is factory preset “close” for 60Hz operation. In accessory kit, there is a special wrench that can be used to operate the open/close status of the valve.

If *60Hz* operation, please ensure the shaft lever of the valve at *CLOSE* position as shown below:



Shaft lever CLOSE status for 60Hz

If *50Hz* power supply, please ensure the shaft lever of the valve at *OPEN* position by means of the special wrench as shown below:



Shaft lever OPEN status for 50Hz

Illustration 13 configuration of the valve for different power frequency

After setting the valve in the correct position, Please *remove the wrench from the valve and preserve it properly* in a location away from the chiller (maybe with the user manual) in order to avoid wrong operation to the valve. Replace the rear panel.

Connect hose and quick disconnect.

1) Fill the reservoir to the indication line on the indication tube with propylene glycol and purified water mixture(50/50).

2) Connect power cord to appropriate terminal block in the GE system Power Distribution Unit.

3) Turn on the power switch located on the front control keyboard.

4) Shift the “stop/start” switch to the start position and the unit will run.

Pump air bleed:

Before the first start-up, it is strongly recommended to bleed air in pump to protect the pump seal from wear-and -tear. Please follow these steps below to finish air bleed.

1) Remove the right side cover, and find the air bleed valve on pump shown in Illustration 13a:

Illustration 13a Pump Air Bleed Valve

2) Use wrentch to loss the nut(smaller one) on top of valve till see liquid flow out of small whole on side of valve (on side of bigger nut), then tigthen the nut and make sure no leak from the side whole.

3) Mounting back the panel removed in step 1.

CAUTION: NEVER USE PUMP TO DRAIN COOLANT IN SYSTEM.

5. Section 5 –Operation

5.1. Control Panel

The MED1COOL/HX is controlled by the control panel located on the front panel.

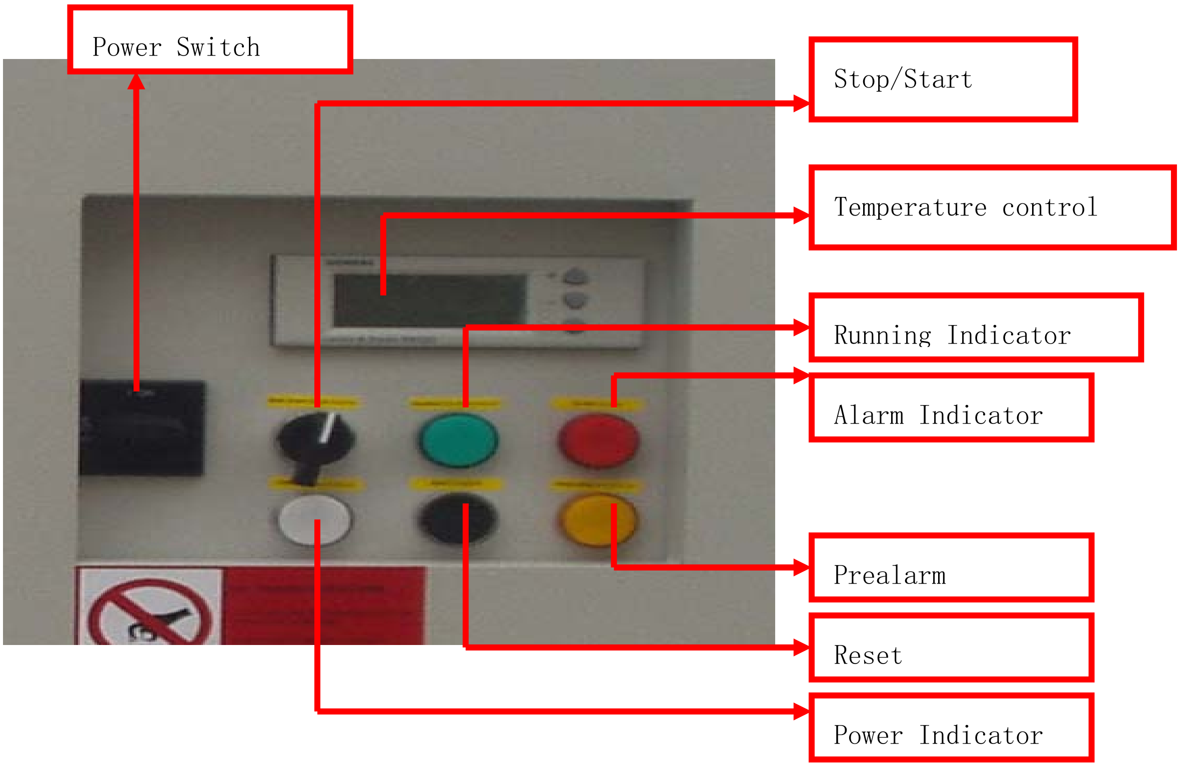


Illustration 14 Control Panel

Power Switch: Power ON / OFF

Power Indicator: Light when power on

Running Indicator: Light when operate normally

Alarm Indicator: Light when reservoir level reaches low alarm line

Pre-alarm Indicator: Light when reservoir level reaches high alarm line

Reset: Push to reset Alarm Indicator

Stop/Start: To start/stop running

The temperature control consists of a LED display and three input keys for the following functions:

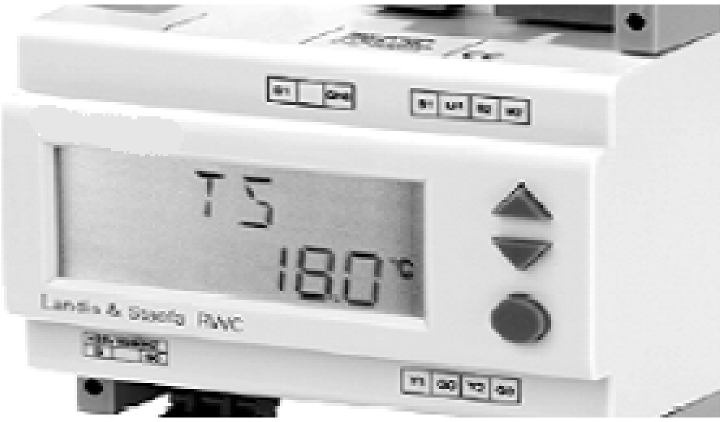


Illustration 15 Temperature Control

The keys work as follows:

● The select button is used to enter or save the value adjustment

▲ The operating buttons are used for viewing and adjusting

▼ parameters.

The setpoint on display can be adjusted by pushing the ● enter/save button, increase value by pressing the ▲ "+" button or decrease the value by pressing the ▼ "–" button, and when the required value is reached, press the ● enter/save button to save the new value.

5.2. Temperature Setting and Adjustment

After setting up and filling the unit:

1)Turn on the power switch on the front control panel. The LED display indicates current fluid temperature. Your required operating temperature can now be set.

***NOTE*: THE EXCHANGER HAS BEEN PRESET ON 20 °C (68°F).**

2)To set day time operation: Press ▲ "+" button once, there shall be " SP-C☼ 20.0°C" displayed. Press ● once, "20.0°C" flashes. Press ▲ "+" or ▼ "-" to the temperature you required. Then press ● to stop the flash. Press ▲ "+" until display return to the normal mode.

3)To set night time operation: Press ▲ "+" button once, there shall be " SP-C(20.0°C" displayed. Press ● once, "20.0°C" flashes. Press ▲ "+" or ▼ "-" to the temperature you required. Then press ● to stop the flash. Press ▲ "+" until display return to the normal mode.

5.3. Draining the Unit

Power off the unit. Follow all applicable G.E. Lock-out, tag-out procedures. Remove screws from the right cover. Open the drain valve allowing unit to drain.

Illustration 16 Draining the Unit

5.4. Alarms

In the event of a low level fluid, the alarm lamp will illuminate. (Refer to page 14).

When fluid gets down to the high alarm level, the "Prealarm" indicator will illuminate. Please check if there is a leakage within the system. Add mixture to reaches "CHARGE LEVEL" line.

When fluid get down to the low alarm level, the "Alarm" indicator will illuminate and flow switch shut down the pump. Check for leakage and add mixture to reaches "CHARGE LEVEL" line. Press "RESET" button to reset alarm before re-start.

6. Section 6 – Troubleshooting

WARNING! Refer servicing to qualified service personnel. Always disconnect the AC power cord before opening the MEDICOOL cover, as electrical shock hazards exist inside.

6.1. Unit Disabled

Check the power to the unit. Be sure the circuit breaker on the front panel is on. If the unit continues to display an error message or no message, request service.

6.2. No Pumping/Pump Failure

Check if pump motor is running (the running indicator light is on when the pump motor is running).

Check if fluid level reach the charge level, Check fluid level in the whole system to be sure the pump can receiving fluid.

Before doing any troubleshooting except mentioned above, the customer should contact GE Service.

CAUTION!

- Never disassemble the MED1COOL/HX unit as irreparable damage may occur.
- Never store or operate the MED1COOL/HX Unit over 55°C or below -34°C.

7. Section 7 – Maintenance

During the warranty period the maintenance check-up interventions are compulsory. Under normal circumstances, the system should be maintained twice a year.

A detailed and periodic visual inspection of the equipment is always important to ensure good operation.

8. Section 8 – Service and Technical Support

If you have followed the troubleshooting steps and your unit fails to operate properly, contact the company where unit was purchased. Have the following information available for the customer service person:

– Model and Serial Number

– Date of purchase and your purchase order number

– Suppliers' order number or invoice number

– A summary of your problem

Customer satisfaction is our highest priority. Please contact us immediately for technical assistance whenever you have questions or concerns. We can be reached in by using the contact numbers at the bottom of each page.

9. Spare Parts

Part	GE Part Number

Heat Exchanger	2343325
Quick Disconnect Kit	2341077-2
Glycol (50/50 mix)	2297672

INDEX OF ILLUSTRATION

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Appendix A: Wiring Diagram

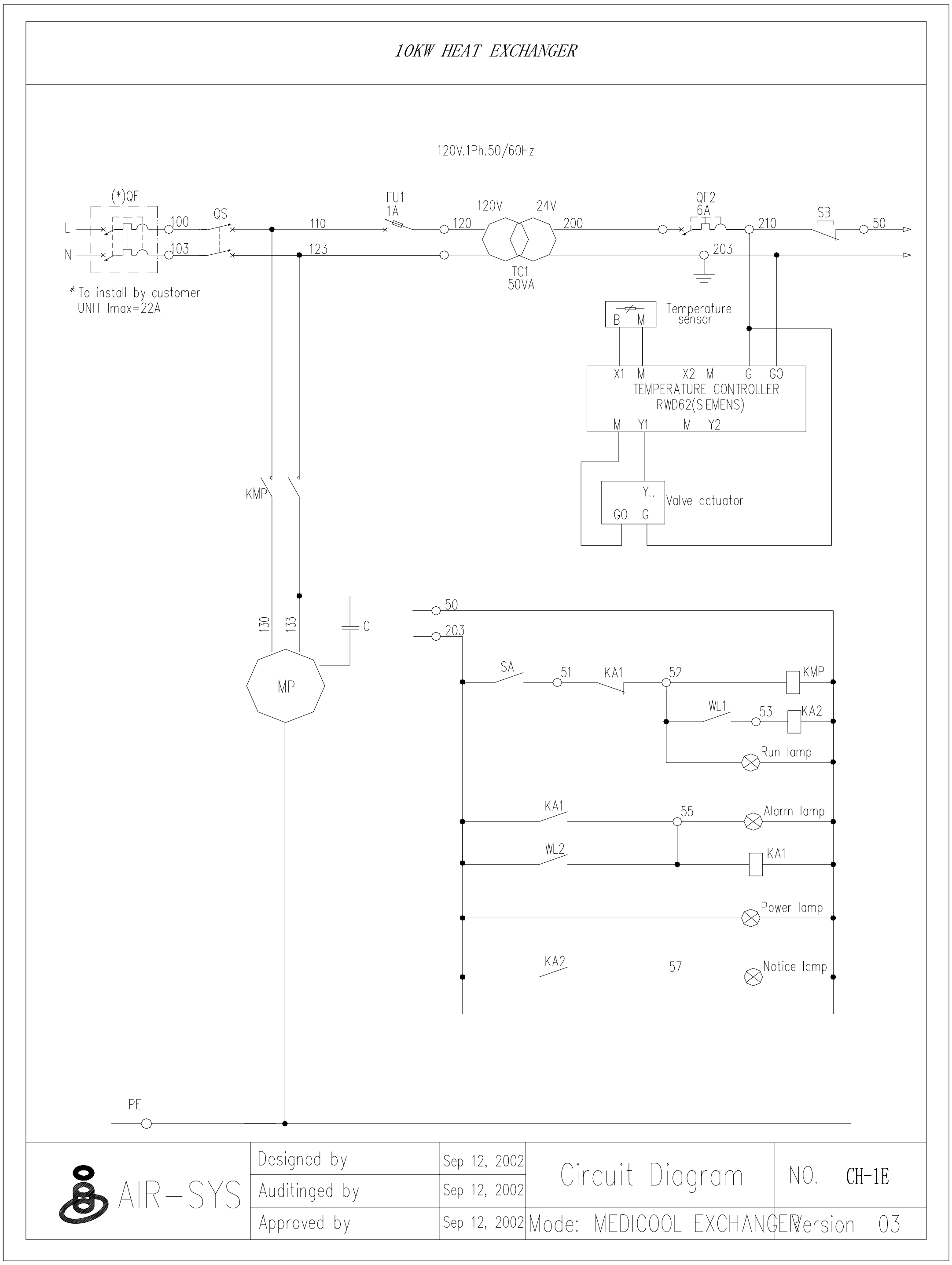
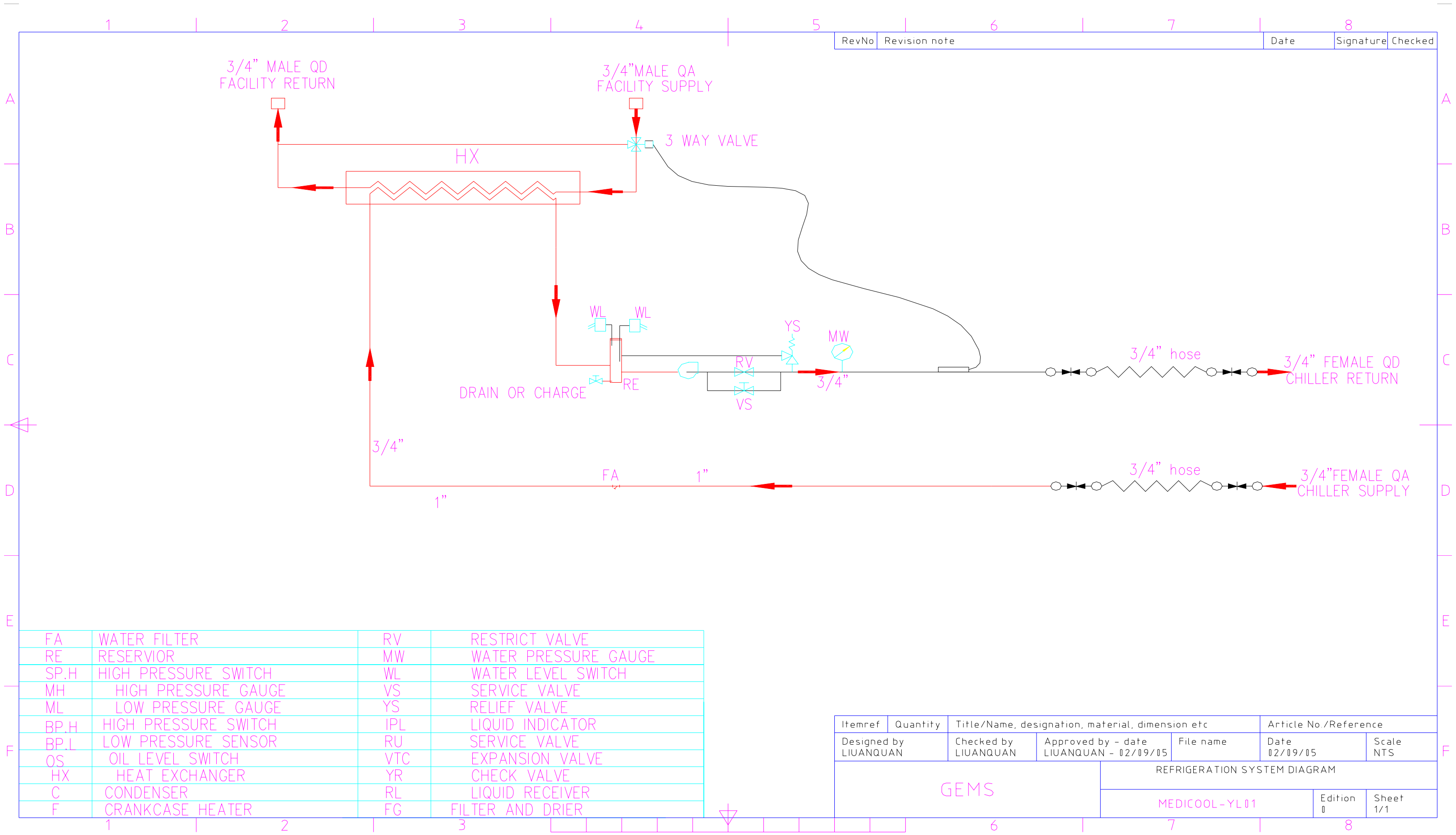


Illustration 177 Wiring Diagram

Appendix B: Pipe diagram



FA	WATER FILTER	RV	RESTRICT VALVE
RE	RESERVIOR	MW	WATER PRESSURE GAUGE
SP.H	HIGH PRESSURE SWITCH	WL	WATER LEVEL SWITCH
MH	HIGH PRESSURE GAUGE	VS	SERVICE VALVE
ML	LOW PRESSURE GAUGE	YS	RELIEF VALVE
BP.H	HIGH PRESSURE SWITCH	IPL	LIQUID INDICATOR
BP.L	LOW PRESSURE SENSOR	RU	SERVICE VALVE
OS	OIL LEVEL SWITCH	VTC	EXPANSION VALVE
HX	HEAT EXCHANGER	YR	CHECK VALVE
C	CONDENSER	RL	LIQUID RECEIVER
F	CRANKCASE HEATER	FG	FILTER AND DRIER

Illustration 18 PIPING DIAGRAM

Appendix C: Material Safety Data Sheet for Propylene Glycol